Mathematics for AI and Machine Learning

Course outline | 36 hours, 18 sessions of 2 hours | Updated 4 November 2026

What it is

The mathematics that sits under machine learning and modern AI, taught the way KLAi teaches everything else. Each idea comes as a picture first, then a small calculation by hand, then the same idea on a computer, then a real use in AI. The course follows the three pillars that the main references share: linear algebra, calculus and optimisation, and probability and statistics. It adds information theory, because that is where the training loss of language models comes from. By the end a learner can read the equations in a machine learning paper or in a library manual, and has built linear regression, logistic regression, PCA, a small neural network and attention from first principles.

Linked to the AI tools people use today

Every session ends by showing where the mathematics appears in tools such as ChatGPT, Gemini and Claude. These are products built on large language models. Their exact designs are not published, so the course uses the published transformer design and open models to show the same mathematics, and it says plainly which parts are published and which are not. Examples: embeddings and cosine similarity in document search, matrix multiplication in every layer, the softmax and temperature that choose the next word, cross-entropy and perplexity as the training score, and attention as the core step.

Who it is for

- **Learners who code.** Analysts, engineers and students. They work in Python (NumPy and Matplotlib in Google Colab, which is free).
- **Learners who do not code.** Career changers and managers. They do the same exercise in Google Sheets or Excel on the same data, so the idea is identical and only the tool differs. Every session has both versions.
- **Entry requirement.** High school algebra. No calculus or statistics needed. Taking it before the Machine Learning course is best, and it also works alongside it.

Dates, fees and enrolment

- First intake starts on 21 November 2026. Enrolment is ongoing, so learners can still sign up after the start date.
- Fee: UGX 1,000,000 per learner, the same as the other KLAi courses. Groups of 1 to 5, group pricing on request, and in-house training for teams.
- Evenings and weekends, in two-hour sessions. Online, at the Innovation Hub in Nakawa, or at the learner's workplace.

How every session runs (2 hours)

Starts	Minutes	What happens
0:00	10	Recall, then a real question from AI that today's mathematics answers
0:10	35	Concept: a picture, then one example calculated by hand
0:45	10	Break
0:55	45	Lab in Python, Google Sheets or Excel: predict, run, then change one thing
1:40	15	Where this appears in ChatGPT, Gemini, Claude and other ML, and an exit check
1:55	5	Save the work

The eighteen sessions

#	Session	What is taught	Where it appears in ChatGPT, Gemini, Claude and other ML
1	Functions, notation and logarithms	Reading sums and functions; exponentials and logarithms	A model scores each possible next word with a probability; training works with the logarithm of that probability
2	Vectors	Data rows as vectors; dot product, length, distance, cosine similarity	Text becomes vectors (embeddings); cosine similarity finds the right passage when AI searches your documents
3	Matrices	Matrices as tables and as transformations; multiplication and shapes	Each layer of a language model is a matrix multiplication; "billions of parameters" means billions of numbers in matrices
4	Solving linear systems	Equations as matrices; inverse, rank, determinant; least squares	Fitting a line is solving equations; why too many features and too few rows break a fit
5	Eigenvalues and eigenvectors	Directions that a matrix only stretches	Where data varies most; the basis of PCA and of understanding embeddings
6	SVD and PCA	Breaking a matrix into simple parts; reducing dimensions	Compression and noise removal; low-rank adaptation (LoRA) fine-tunes big models by training small matrices
7	Derivatives	Slope of a curve; rules; checking with numbers	The slope of the loss says how wrong the model is and which way to change it
8	Partial derivatives and the gradient	Many inputs; the direction of steepest increase	A model with billions of parameters has a gradient with billions of entries
9	The chain rule and the Jacobian	Derivatives of functions of functions; computation graphs	Backpropagation through dozens of layers is the chain rule used again and again
10	Gradient descent	Learning rate, stochastic gradient descent, momentum, Adam, convexity	How ChatGPT-style models are trained: an Adam-type optimiser with a learning-rate schedule
11	Backpropagation by hand	A tiny network forward and backward, then in matrix form	What happens in every training step, and why training costs far more computing than answering a question
12	Optimisation in practice	Regularisation, constraints in plain terms, overflow and the stable softmax	Why long outputs do not overflow, why many models use 16-bit numbers, and how overfitting is controlled
13	Probability basics	Random variables, expectation, variance, covariance	A chat model outputs a probability for every possible next word
14	Distributions and sampling	Bernoulli, binomial, Gaussian, multivariate Gaussian; the central limit theorem; confidence intervals and A/B tests	Temperature and top-p sampling pick the next word, which is why the same question can give different answers; how to test whether a new AI feature really is better
15	Conditional probability and Bayes	Updating beliefs with evidence; a small Naive Bayes classifier	Spam filters and diagnosis; reasoning with uncertain answers
16	Maximum likelihood and MAP	Why squared error and cross-entropy are the right losses; bias and variance	Training a language model is maximum likelihood; weight decay is a MAP prior
17	Information theory	Entropy, cross-entropy, KL divergence, softmax, perplexity	Cross-entropy is the training loss of language models, and perplexity is their standard score
18	Capstone	Build linear regression, logistic regression, PCA, a small network and attention from first principles	Attention is the core step inside ChatGPT, Gemini and Claude; the build is individual and auto-graded

Kept out on purpose

Measure theory, formal proofs, convex duality, tensor calculus and the finer points of numerical linear algebra. They are useful for researchers but lose most learners. Derivations appear only where they feed something the learner then implements: the least-squares solution, backpropagation, and the gradient of softmax with cross-entropy.

Assessment and what learners keep

- No exams. Each session ends with an automatic check of the notebook or spreadsheet, the same kind of check KLAi already uses in the IUEA course.
- The capstone (session 18) is an individual, auto-graded build. The Python version is a notebook, and the spreadsheet version is a guided workbook (Google Sheets or Excel) with the same numbers.
- A certificate at the end, as for every KLAi course.
- A one-page formula sheet that each learner writes, with every formula named in plain words and in the industry term.

Decisions already made

- 36 hours in 18 sessions. Both coding and non-coding learners. Preferably taken before the Machine Learning course.
- High school algebra as the only entry requirement. Derivations only where they feed an implementation.
- Start 21 November 2026 with ongoing enrolment. The fee is the same as the other courses. The non-coding track works in Google Sheets and in Excel.

Next step

The flyer comes first, so that you can share and market the course. After you approve the flyer, I build the rest in the style of the Applied AI course: KLAi slides with spoken notes, Python notebooks, spreadsheet workbooks for Google Sheets and Excel, automatic checks, and a facilitator guide.